

Introduction to Vectors

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Linear algebra begins with two operations on vectors: addition and multiplication by numbers. Combining them produces linear combinations, which may fill a line, a plane, or a higher-dimensional space. The remarkable feature is that the same geometric picture remains correct even when the dimension is too high to draw.

Adapted from Gilbert Strang, Chapter 1 introduction, p. 1.

VECTORS AND LINEAR COMBINATIONS

$$v + w = (v_1 + w_1, \dots, v_n + w_n), \quad cv = (cv_1, \dots, cv_n).$$

Linear combination and span:

For $v_1, \dots, v_n \in \mathbb{R}^m$ and $c_1, \dots, c_n \in \mathbb{R}$, the vector $\sum_{j=1}^n c_j v_j$ is a linear combination. Its possible values form

$$\text{span}\{v_1, \dots, v_n\} = \left\{ \sum_{j=1}^n c_j v_j : c_j \in \mathbb{R} \right\}.$$

Matrix form:

If $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$ and $x = (x_1, \dots, x_n)^T$, then x_j is the coefficient of the j th column a_j :

$$Ax = \sum_{j=1}^n x_j a_j = b.$$

Linear dependence theorem:

$v_1, \dots, v_n$ are dependent iff $\sum_{j=1}^n c_j v_j = 0$ for coefficients $c_j \in \mathbb{R}$ that are not all zero.

DOT PRODUCTS AND GEOMETRY

$$v \cdot w = v^T w = \sum_{i=1}^n v_i w_i, \quad \|v\| = \sqrt{v \cdot v}.$$

Cosine formula:

for nonzero v, w , $\cos \theta = \frac{v \cdot w}{\|v\| \|w\|}$.

Cauchy-Schwarz inequality:

$$|v \cdot w| \leq \|v\| \|w\|.$$

Triangle inequality:

$$\|v + w\| \leq \|v\| + \|w\|.$$

Pythagorean theorem:

$$v \cdot w = 0 \Rightarrow \|v + w\|^2 = \|v\|^2 + \|w\|^2.$$

$$\|v - w\|^2 = \|v\|^2 - 2v \cdot w + \|w\|^2.$$

MATRICES AND EQUATIONS

Invertibility criterion (square case):

A is invertible iff $Ax = 0$ has only $x = 0$ iff its columns are independent; then $x = A^{-1}b$.

$$AB = (Ab_1 \cdots Ab_p), \quad (AB)^T = B^T A^T, \quad A(I) = A.$$

PROBLEM SET

Q1.

Let $v = (v_1, v_2)$ and $w = (w_1, w_2)$ be nonzero vectors with $v_1 w_1 \neq 0$. Suppose the product of the slopes of the lines joining the origin to v and w is -1 . Show that $v \cdot w = 0$, and hence that the vectors are perpendicular.

Q2.

- (i) Let $A \in \mathbb{R}^{m \times n}$ and $x \in \mathbb{R}^n$. For each i , let $r_i \in \mathbb{R}^n$ be the column vector whose transpose r_i^T is the i th row of A . If $Ax = 0$, show that $r_i \cdot x = 0$ for every i .
- (ii) Hence explain why, when the rows span a plane in $\mathbb{R}^3$, that plane is perpendicular to x .

Q3.

- (i) For

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix},$$

solve $Ax = b$ for an arbitrary $b = (b_1, b_2, b_3)^T$.

- (ii) State the inverse matrix A^{-1} .

Q4.

- (i) Find two different combinations of the three vectors $u = (1, 3)$, $v = (2, 7)$, and $w = (1, 5)$ that produce $b = (0, 1)$.
- (ii) Show that, for any three vectors in the plane, if one combination produces b , then there are infinitely many distinct linear combinations that produce b .

Q5.

Let x, y, z be real numbers, not all zero, satisfying $x + y + z = 0$. Set $v = (x, y, z)$ and $w = (z, x, y)$. Find the angle between v and w by showing that

$$\frac{v \cdot w}{\|v\| \|w\|} = -\frac{1}{2}.$$

Q6.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 real matrix. Prove that the rows of A are linearly dependent if and only if the columns of A are linearly dependent.

Q7.

- (i) Show that there cannot be three nonzero, mutually orthogonal vectors in $\mathbb{R}^2$.
- (ii) Show that there are at most n nonzero, mutually orthogonal vectors in $\mathbb{R}^n$.
- (iii) *Optional.* Fix a small angle $\varepsilon > 0$. Make a conjecture about the growth, as n increases, of the maximum number of unit vectors that can be placed in $\mathbb{R}^n$ such that every pair forms an acute angle between $90^\circ - \varepsilon$ and 90° .

Q8.

- (i) Let u and v be independent unit vectors chosen uniformly from the unit circle in $\mathbb{R}^2$. Find the mean value of $|u \cdot v|$.
- (ii) *Optional.* Repeat the calculation for independent unit vectors chosen uniformly from the unit sphere in $\mathbb{R}^3$. Does the comparison support your conjecture in Q7(iii)?

Hint: If a continuous random variable X has probability density function f_X , then

$$\mathbb{E}[g(X)] = \int g(x) f_X(x) dx.$$

By rotational symmetry, one vector may be fixed; determine the distribution of the dot product using spherical surface area.